YOSEF D. STIFANOS

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EDUCATION

University of Illinois at Urbana-Champaign — **Urbana, Illinois**
M.S. in Cyber and Information Management — *August 2026 – Expected May 2028*
- *Illinois Cyber Security Scholars Program, NSF CyberAI/CyberCorps, Information Trust Institute*

University of Illinois at Urbana-Champaign — **Urbana, Illinois**
*B.S. in Information Sciences + Data Science & **Cybersecurity*** — *August 2022 - May 2026*
- **GPA:** 3.76/4.00, *6x Dean's List, NSBE Chicago Scholarship Recipient*
- **Relevant Coursework:** Data Science 1&2, Machine Learning, Risk & Cybersecurity, Fundamentals of Information Security, Privacy & Information Technology, Database Concepts and Applications, Cybersecurity Concepts & Practices

EXPERIENCE

Caterpillar Inc. — **Irving, Texas**
Cybersecurity Engineer — *March 2026 – August 2026*
- Cut manual reconciliation across compliance data formats, as measured by faster, standardized control catalog outputs, by building Python automation to transform data across Excel, JSON, CSV, and OSCAL.
- Identified a multimillion-dollar cost savings opportunity and achieved 100%+ of project goals across all 7 deliverables by co-developing a multi-agent AI platform leveraging PDF requirement extraction, Azure AI Search-powered component discovery, automated BoM generation, and an engineer-facing chatbot, reducing BoM creation from hours to minutes and saving 15+ hours per week per design team with delivering $100k+ in estimate annual value.

Freddie Mac — **McLean, Virginia**
Technology Analyst — *May 2025 – July 2025*
- Participated in planning and executing 2 tabletop exercises simulating cyber incidents, system outages, and recovery scenarios to stress-test enterprise resilience and incident response readiness.
- Mapped recovery dependencies across hundreds of enterprise applications, identifying critical systems and post-incident prioritization gaps to support business continuity planning.
- Produced a Power BI report for the Cyber/Technology Resilience team, synthesizing attack path analysis and resilience gap findings into actionable executive deliverables.

The Grainger College of Engineering — **Urbana, Illinois**
Network & Infrastructure Engineering Student Consultant — *April 2025 – March 2026*
- Provided AV/IT support across 100+ Grainger Engineering classrooms, resolving ~100 tickets over two semesters to maintain high-availability instructional environments.
- Administered ~150 IP address allocations using IPAM with zero documentation errors, ensuring accurate network records and reliable enterprise connectivity.

Google — **Remote, California**
Cybersecurity Scholar — *August 2025 – December 2025*

Vivacity Tech PBC — **Remote, St. Paul, Minnesota**
Software Quality Engineer Intern — *January 2024 – June 2024*

University of Illinois Technology Services — **Urbana, Illinois**
Information Technology Operations Analyst — *September 2022 – February 2025*

PROJECTS

NIST Password Generator + Passphrase Tool — **Urbana, Illinois**
Independent Owner — *July 2025*
- Integrated the Have I Been Pwned API to detect compromised passwords in real time using k-anonymity, ensuring plaintext credentials are never exposed.
- Leveraged Claude AI to validate cryptographic randomness logic and support rigorous security testing, ensuring implementation correctness against NIST SP 800-63B standards.
- **Tools**: HTML, JavaScript, Web Crypto API, Have I Been Pwned API, Cybersecurity Standards

Predictive Modeling on Game Sales and Playtime — **Urbana, Illinois**
Group Lead — *December 2024*
- Developed regression and classification models to forecast game sales and playtime behavior.
- Engineered features, trained/tested models, and evaluated performance using R^2, AUC, and RMSE.

ACTIVITIES AND LEADERSHIP

NSBE Member – Coordinated STEM workshops for local schools, increasing outreach by 30%.
SIGPwny – Participated in weekly red/blue security workshops and CTF competitions focused on real-world hacking techniques.

SKILLS

Languages: Python, JavaScript, HTML, CSS, SQL
Tools: Git, Jira, Jupyter Notebooks, Korbyt, MySQL, MongoDB, Wireshark, IPAM, Splunk, Windows, Linux, MacOS, LLM API Integration, Azure AI, Prompt Engineering
Frameworks/Libraries: Pandas, NumPy, Matplotlib, Scikit-learn
Certifications: Google Cybersecurity Professional Certificate
Cybersecurity: CIA Triad, OSI Model, MITRE ATT&CK, ServiceNow IRM, Kali Linux, NMAP, SIEM, OSCAL, SBOM, Vulnerability Assessment, Patch Management, System Security Assurance